Before It Breaks

Network-aware reliability and contingency planning for UW–Madison

THE CAP TIMES

UW-Madison's summer air conditioning woes cost $15.8 million

WISCONSIN PUBLIC RADIO

More than 20 buildings closed on UW-Madison campus because of air conditioning issues

15 WMTV

Board of Regents approved millions for aging infrastructure weeks before UW-Madison AC issues, records show

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Total incident cost

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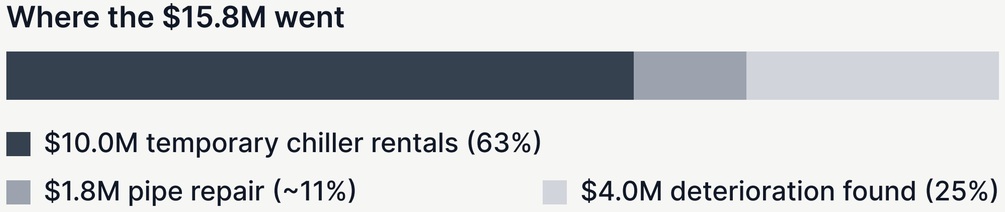
$15.8M

Total incident cost

63%

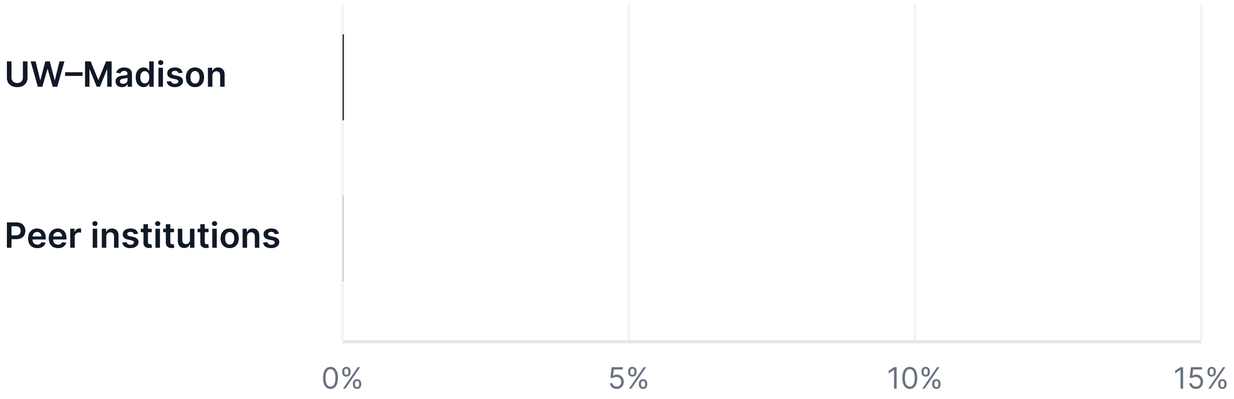
of the total bill was spent renting temporary chillers

About 30 temporary chillers ran until every building reopened on Aug 10 — 54 days after the break.



The backlog demands better prioritization.

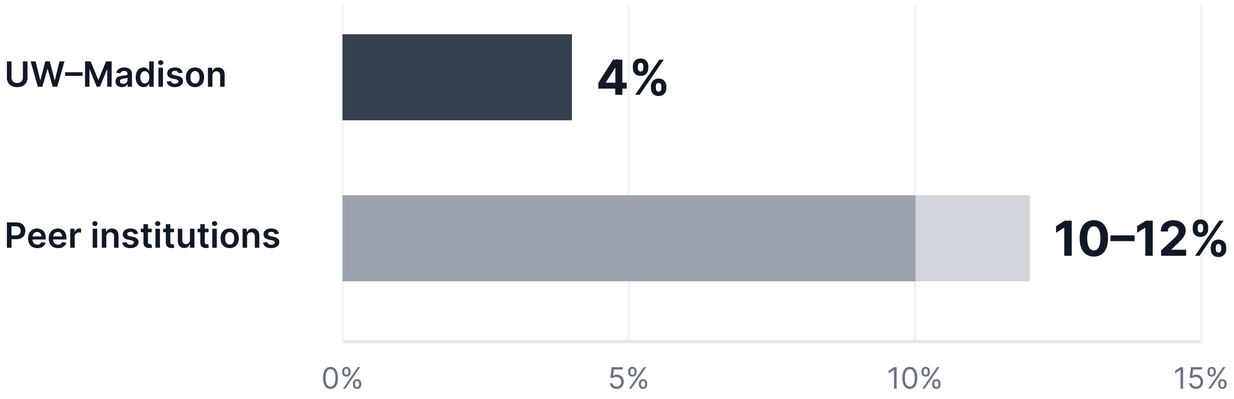
Preventive maintenance share of the O&M budget



Peers: Illinois, Minnesota, Ohio State. Light segment marks the 10–12% range. Axis starts at 0%.

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$1.5B

deferred-maintenance backlog

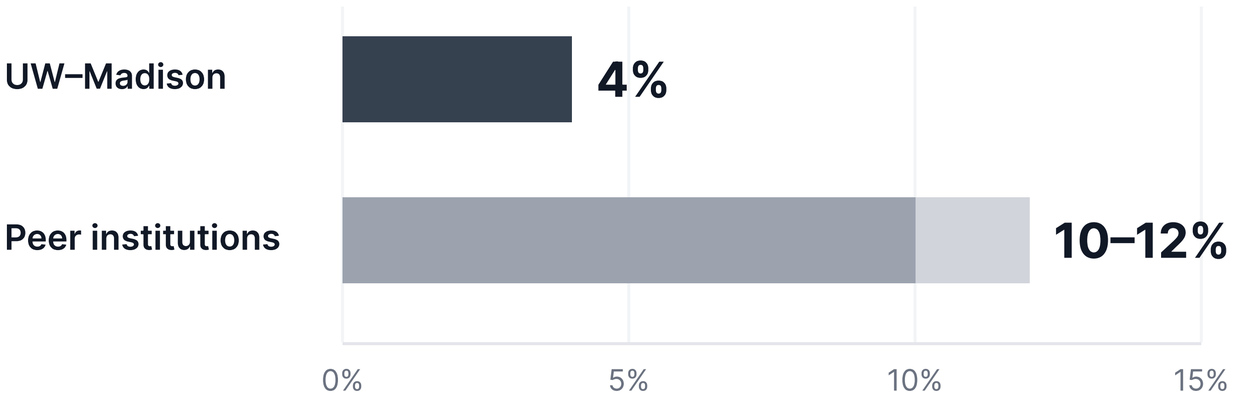
WPR, Feb 2019 · historical context

120

facilities emergency calls in May 2021

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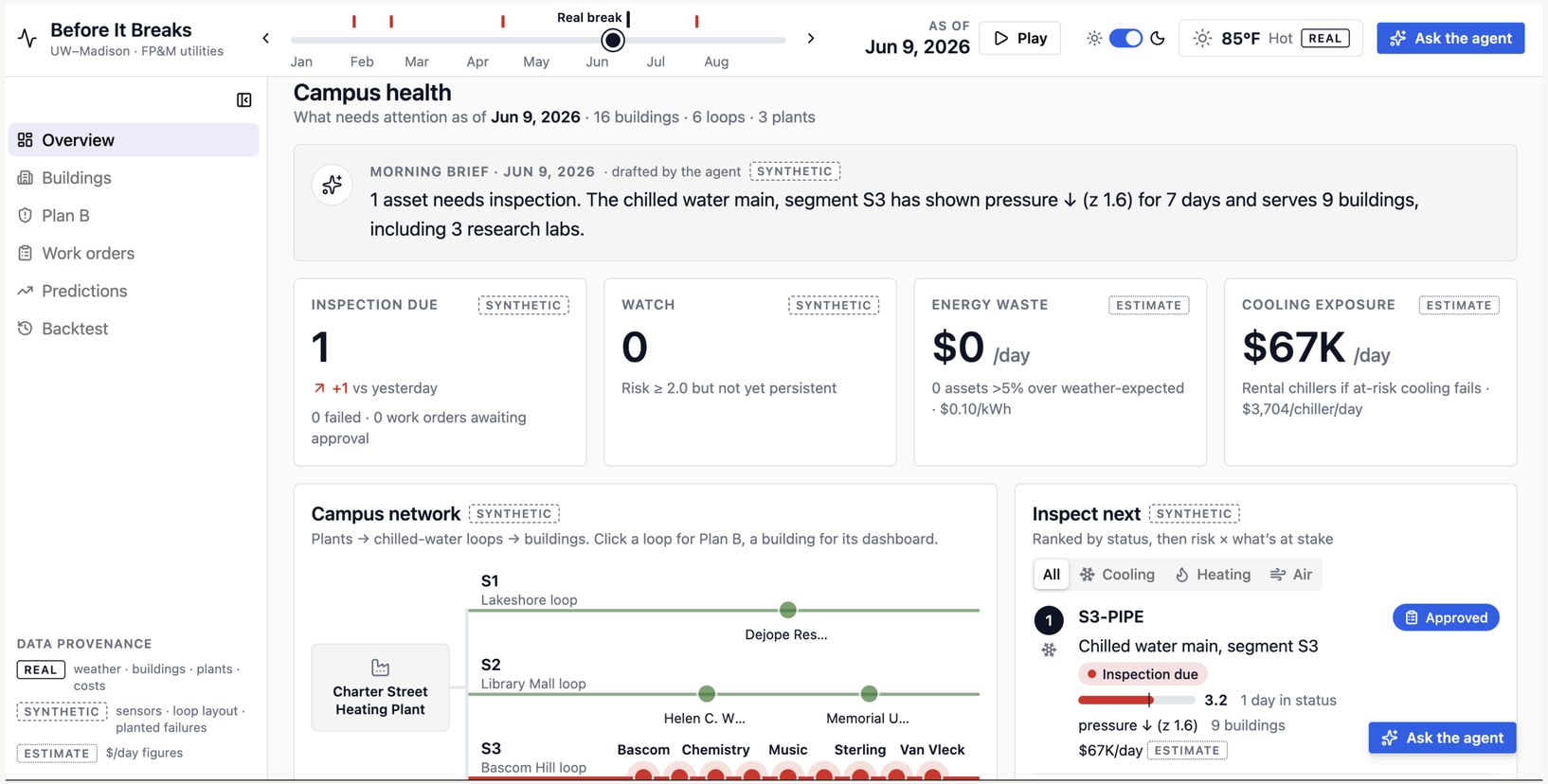
facilities emergency calls in May 2021

Connect equipment condition, campus impact, and a prepared response.

LIVE DEMO

From warning to approval and an outage plan.

June 17 reenactment



01 Prioritize
Morning brief and 'Inspect next': status, then risk × what's at stake

02 Explain
Asset detail: energy vs. weather-expected, pressure and flow drivers

03 Approve
Agent-drafted work orders; a facilities manager approves or dismisses

04 Simulate
Outage plan: fail a chilled-water loop; stage rental chillers by criticality

05 Evaluate
Backtest: 5 planted failures, 4 decoys, adjustable threshold

Resilience during failures. Efficiency every day.

Prepared contingency

$3.5M–$4.5M

Potential rental savings in a June-scale scenario

Assumption: 35–45% reduction against a $10M rental baseline.

Resilience during failures. Efficiency every day.

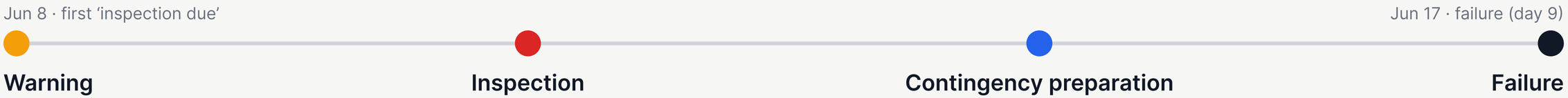
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9-DAY WARNING · S3 REENACTMENT



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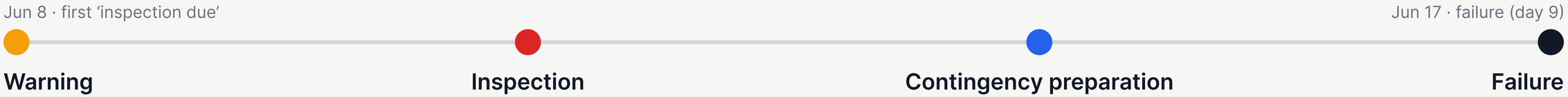
Everyday energy

$180K–$320K / year

Illustrative savings across 50 assets

Modeled scenario; degrading assets use 15–30% excess energy.

9-DAY WARNING · S3 REENACTMENT



Built on C3. Designed for connected infrastructure.

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C3 data model

Assets, utility topology, building criticality, telemetry

→

Feature & risk pipeline

Weather-normalized baselines and persistent-anomaly scores

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Agent recommendations + human approval

Ranked inspections, draft work orders, outage plans

→

React operations interface

Queue, asset detail, approval, backtest

Connect

Existing building-automation and work-order data. Uses existing telemetry where suitable signals are available.

Validate

On historical telemetry and work-order outcomes before any operational deployment.

Adapt

Topology and asset models for hospitals, campuses, and district-energy networks.

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Topology and asset models for hospitals, campuses, and district-energy networks.

Fix it before it breaks. Have the outage plan ready.

How an asset gets flagged.

01 **Its own baseline**

Each asset is compared with its own normal behavior, not a fleet-wide fixed threshold.

02 **Weather-normalized energy**

Expected energy is adjusted for outdoor conditions before excess is measured.

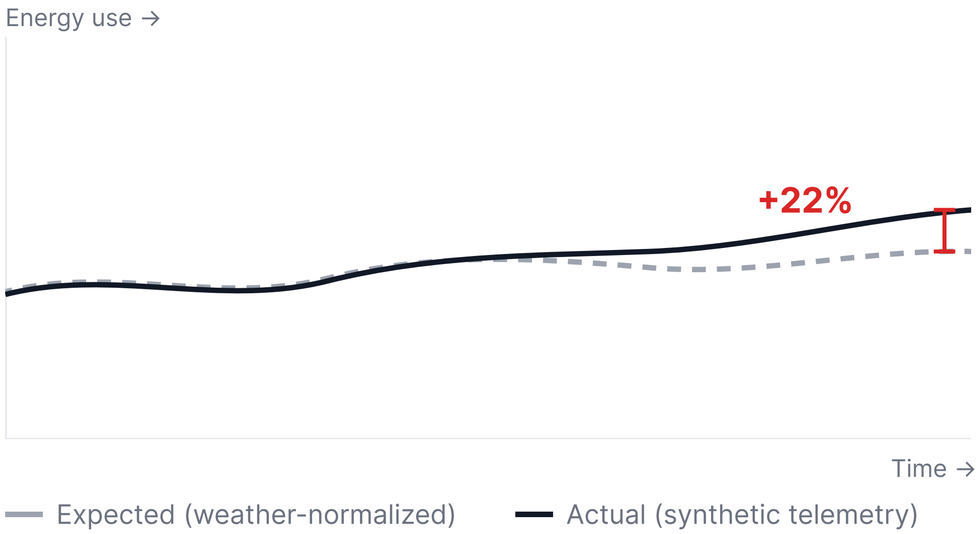
03 **Persistence**

Pressure, flow, vibration, and energy anomalies must persist — single spikes are ignored.

04 **Network impact**

Priority weighs downstream building criticality on the campus utility topology.

Vilas Hall unit heater · energy vs. weather-normalized baseline



Risk score = anomaly score, not a calibrated failure probability.

Backtest: planted failures, decoys, threshold.

Planted failure

Five synthetic failures across heating, air handling, and cooling — including the June 17 reenactment on S3-PIPE.

Decoy scenario

Look-alikes that should not trigger inspection: one-day vibration spike, 5-day sensor dropout, real July heat wave, slow normal aging.

Lead time

Days from first ‘inspection due’ to the planted failure. At threshold 3.0: mean 6.8 days, minimum 6 (synthetic replay).

Alert threshold

Inspection due = risk ≥ threshold on 2 consecutive days (default 3.0). Moving it trades warning time against false alarms.

Threshold-dependent false-alarm tradeoff



Lower threshold
Earlier warnings, more false alarms

Higher threshold
Fewer alarms, later or missed warnings

IN THE LIVE APP

At the default threshold (3.0): 0 of 5 failures missed · 6.8-day mean warning · 2 false-alarm days.

Synthetic replay results; they update as the threshold moves.

What’s real, synthetic, and estimated.

REAL

Historical facts · need source attribution

- June 17, 2026 Charter Street chilled-water main rupture (~4 ft main, ~20 ft deep)
- 23 buildings closed; 11 partially closed; ~30 temporary chillers; all reopened Aug 10
- $15.8M total · $10.0M chiller rentals (63%) · $1.8M pipe repair (~11%)
- $1.5B deferred-maintenance backlog; $21M/yr to stop growth (2019 context)
- Prevention share: UW 4% vs peers 10–12% (historical)
- May 2021 ops report: 2,218 PMs generated, 460 follow-up repairs, 120 emergency calls
- Daily weather (NOAA GHCN-Daily) and building and plant facts (UW–Madison public records)

SYNTHETIC

Demo telemetry, planted failures, replays

- Sensor telemetry: pressure, flow, vibration, energy
- Planted failures and decoy scenarios
- Vilas Hall unit heater: +22% vs weather-normalized baseline
- S3-PIPE flow imbalance, first flagged Jun 8 (reenactment)
- Backtest at threshold 3.0: 6.8-day mean warning, 2 false-alarm days

ESTIMATE

Derived costs, savings, and assumptions

- ~$111K/day rental burn rate (our estimate)
- 35–45% assumed rental reduction → $3.5M–$4.5M
- Degrading assets: 15–30% excess energy
- $180K–$320K/yr across an illustrative 50-asset portfolio

Savings assumptions and arithmetic.

PREPARED CONTINGENCY

$10.0M × 35% = $3.5M

$10.0M × 45% = $4.5M

$10.0M	Historical chiller-rental cost, June 2026 incident
35–45%	Assumed reduction through targeted preparation (assumption, not observed)
$3.5M–$4.5M	Potential rental savings in a June-scale scenario

EVERYDAY ENERGY

$180K–$320K / year

across an illustrative 50-asset portfolio

Inputs needed to validate

- [Asset load or rated capacity]
- [Annual runtime hours]
- [Energy tariff, $/kWh or $/therm]
- [Baseline consumption per asset]
- 15–30% excess energy — modeled
- 50 assets — illustrative portfolio

Arithmetic for this range was not supplied, so inputs are marked pending rather than invented.

Architecture and implementation evidence.

Inputs

- Weather data (real)
- Building facts (real)
- Sensor telemetry (synthetic)
- Utility topology & criticality

→

C3 data model

- Assets, buildings, utility segments, readings
- [Confirm C3 type names]

→

Feature & risk pipeline

- Weather-normalized baselines
- Persistent-anomaly scoring
- Network-impact weighting

→

Agent + human approval

- Ranked inspections
- Draft work orders → approval
- Failure simulation & outage plan

→

React operations UI

- Queue · asset detail
- Approval · outage plan
- Backtest & threshold

IMPLEMENTATION EVIDENCE

- C3 package: pkg22bc8209
- Endpoints: [verify count before citing]
- Repository / demo link: [add]

KNOWN LIMITS

- Telemetry and failures are synthetic
- Risk score is an anomaly score, not a calibrated probability
- No UW deployment; validation on historical telemetry and work-order outcomes is next

The 2026 outage, and where the app fits.

WHEN	WHAT HAPPENED · REAL	WHERE THE APP FITS
Jun 17, 2026	A chilled-water line breaks near the Charter Street plant — a 4-ft pipe, 20 ft underground. Cap Times, Oct 8, 2026	S3-PIPE reenactment: first flagged Jun 8, nine days earlier (synthetic).
Jun 29–30	As dangerous heat arrives, 23 buildings close and 11 partially close; cooling goes to patient care and research. WMTV · Channel 3000, Jun 29, 2026	Outage plan: who loses cooling, with labs and patient care first.
Late June	About 30 temporary chillers are installed; repairs are expected to take at least a month. Channel 3000 · WMTV, Jun 29, 2026	Rental chillers sized and staged per building before a failure (estimate).
Aug 10	All closed and partially closed buildings reopen — 54 days after the break. UW–Madison News	Cooling exposure in $/day while a loop is down (estimate).
Oct 8	Cost revealed: $15.8M, about $10M of it temporary chillers. Cause still under investigation; Regents shift $40M to a related utilities project. Cap Times, Oct 8, 2026	Modeled $3.5M–$4.5M rental savings with a prepared plan (estimate).
2019 context	Backlog about $1.5B; prevention ~4% of O&M vs 10–12% at peers; a polar vortex hits 28 facilities, including Chemistry and Vilas. WPR, Feb 8, 2019	Rank by condition × network impact; compare each asset with its own normal.

Sources and Q&A.

SOURCES

Cap Times · Oct 8, 2026
Cost: $15.8M (~$10M chillers, ~$1.8M line repair, ~$4M deterioration); 4-ft pipe, 20 ft deep; cause under investigation

WMTV and Channel 3000 · Jun 29, 2026
23 buildings closed, 11 partially closed; ~30 temporary chillers; patient care and research prioritized

UW–Madison News · Campus cooling issues
All closed and partially closed buildings reopened Aug 10, 2026

WPR · Feb 8, 2019
$1.5B backlog; $21M/yr to stop growth; preventive maintenance 4% vs 10–12% at peers

FP&M Physical Plant Operations Report · May 2021
2,218 PMs generated; 460 follow-up repairs; 120 facilities emergency calls

NOAA GHCN-Daily · UW–Madison public records
Daily weather (Madison Dane County Regional Airport); buildings and plants

Would this have predicted the actual June 17 incident?

We do not know. The demonstration uses synthetic signals and planted failures. It shows the workflow when detectable deterioration is present.

Why more than threshold alarms?

We compare assets against their own normal, adjust energy for weather, and require persistent anomalies. The backtest exposes the tradeoffs.

What is distinctive?

We combine equipment condition with network impact and prepare a contingency response before failure.